System Requirements:

This guide requires at least **12GB Vram or Higher**. I have not tested it with 8GB VRAM configurations.

Tested Hardware:

- RTX 3060 12GB - Training works but is noticeably slower
- RTX 5070 Ti 16GB - Smooth training experience with faster processing times

The performance difference between 12GB and 16GB VRAM is expected and primarily affects training speed rather than capability.

Note1: This guide does not cover LoRA training for Wan 2.2 While it's technically possible using AIToolkit, it demands high VRAM. Since you'll be using **RunPod** for Wan 2.2, it's far better to follow the video tutorial—it's simpler, clearer, and built for that setup.

- **Wan 2.2 Training Guide: Click Here**

Note2: If you wanna use Ai-toolkit in Runpod you can use my colab notebook: **ClickHere**

This article may be a bit lengthy, but rest assured, it is well worth your time. I will also assume that you are already familiar with the process of installing Ai-ToolKit locally.

If you don't know how to install:

- Install the prerequisites: **Git**, **Python 3.11+**, and **Node.js v22+**
- Clone the repo: `git clone https://github.com/ostris/ai-toolkit.git`
- Install Python dependencies: `pip install -r requirements.txt`
- Navigate to the UI folder and install Node dependencies: `cd ui` → `npm install`
- Start the UI: `npm run dev`

STEP 1: Dataset Preparation

Images: **60+ recommended, 20 minimum** — clear, sharp, varied angles/lighting

Part A — JoyCaption Setup (optimized version by me) for Image Caption

```
git clone https://github.com/official-imvoiid/Joycaption
cd Joycaption
```

Run in order: `GetConda.bat` → `SetEnv.bat` → `InstallRequirements.bat` → `StartTextCaptioner.bat`

4. Bulk Tab, select all images
5. Type/Select what you want the model to tag/focus on
6. Run captioning
7. Download the caption ZIP at the end
8. Unzip it → move all `.txt` files into the **same folder as your images**

Part B — Trigger Word Adder

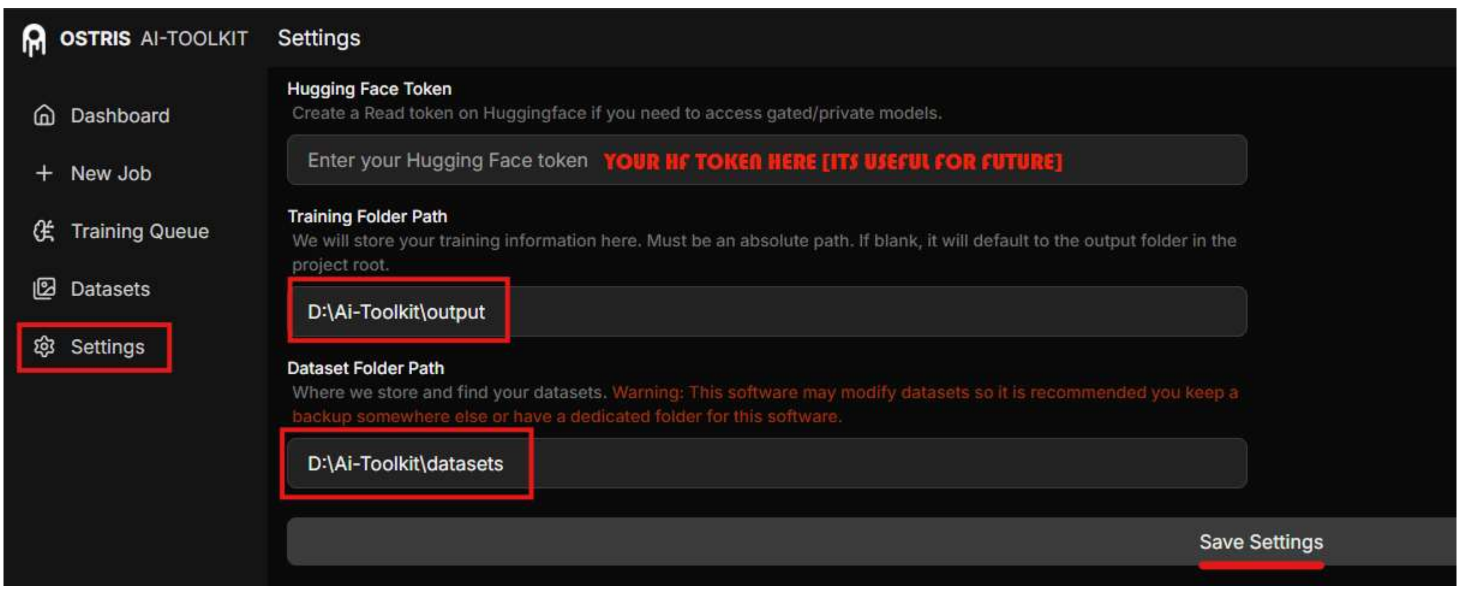
```
https://github.com/official-imvoiid/Random/blob/main/TriggerWordAdder.py
```

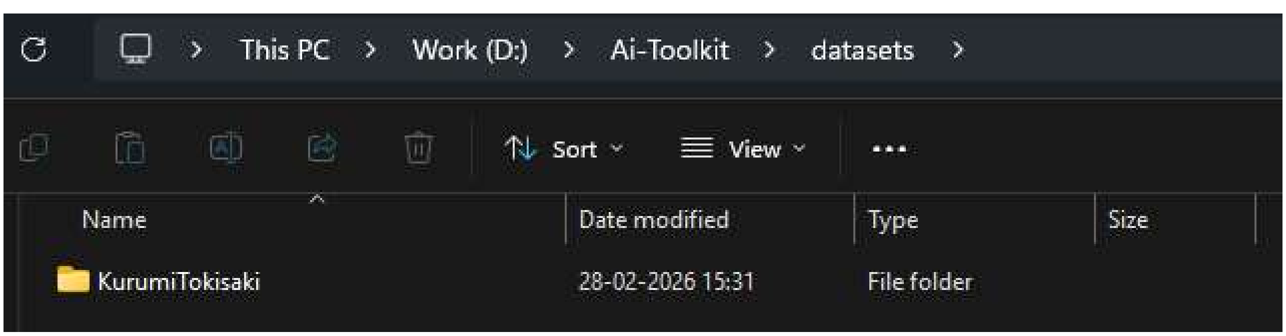
No pip install needed — pure Python. Just run `TriggerWordAdder.py` and paste the path of you folder having images and captions/tags ⚠️ Critical: Type trigger word as `voiid,` (with comma+space)

- ✅ `voiid, other-tags`
- ❌ `voiidother-tags`

STEP 2: Pre-Requisite

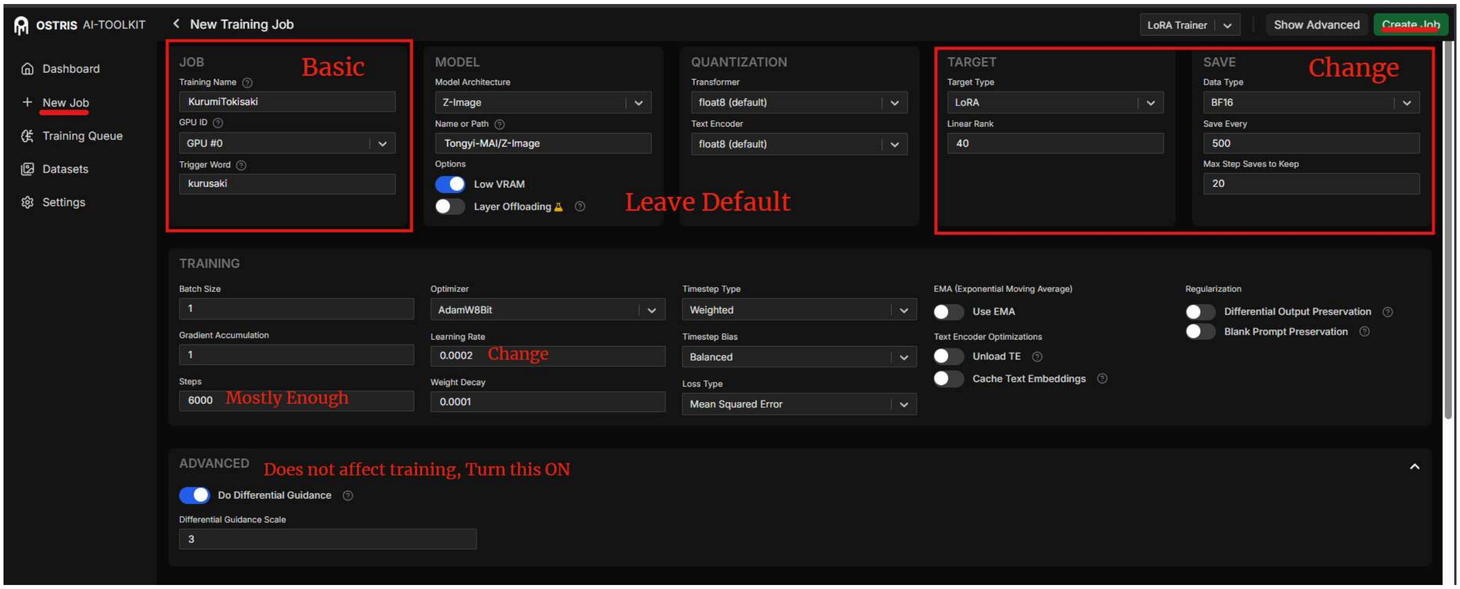
Make Sure to move this folder containing images & caption to <Path-To-Your-Ai-Toolkit>\datasets **And also to apply following settings**

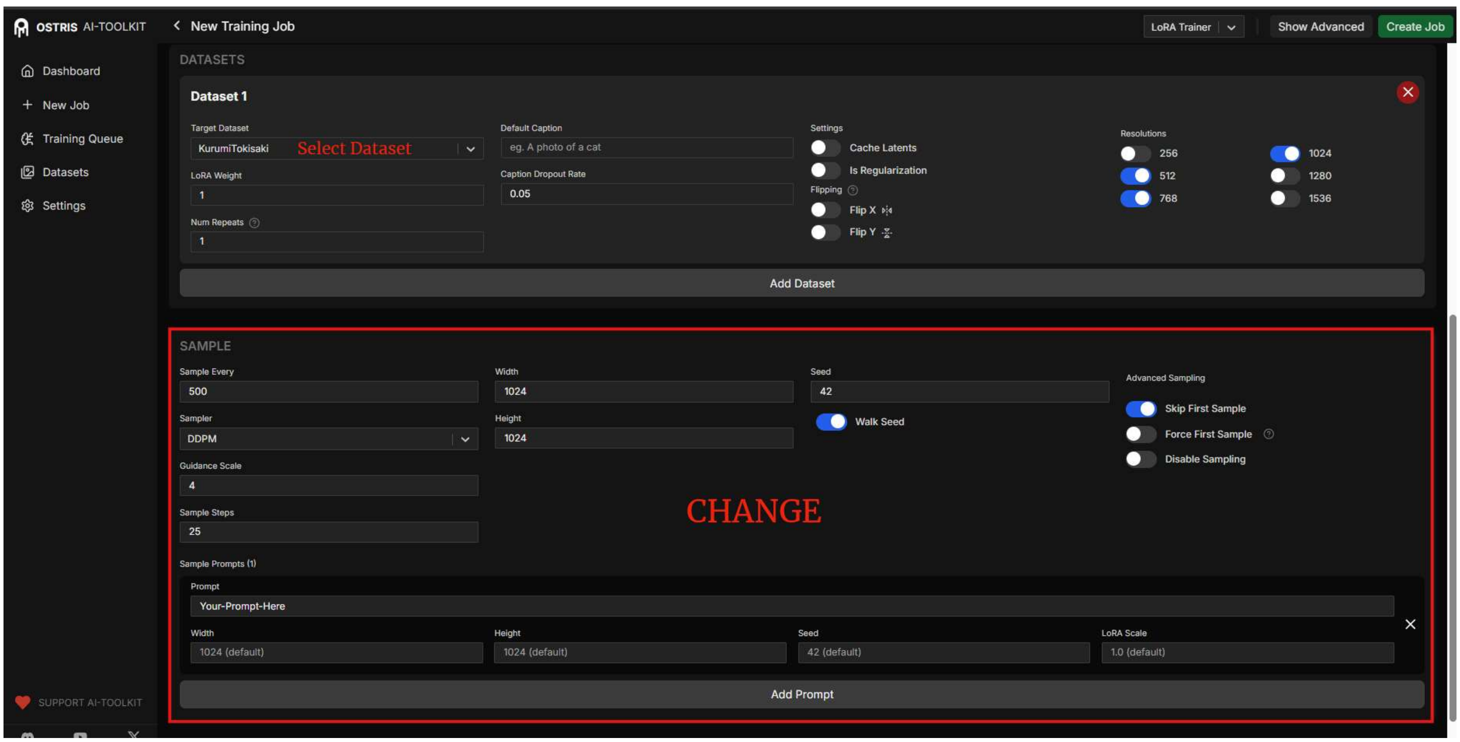




STEP 3: Actual Training

From Here the real training process begins, see these settings [also given as images in attachments]





Click **New Job → LoRA Trainer**

JOB Section

- Training Name = Your LoRA name
- Trigger Word = Same word used in your captions

MODEL Section *(leave default)*

- Architecture: Z-Image
- Model: Tongyi-MAI/Z-Image

- It will auto-download the ~30GB base model to your `.huggingface\hub` folder on first run

QUANTIZATION Section *(leave default)*

TARGET Section *(change)*

- Linear Rank: 40

SAVE Section *(change)*

- Data Type: BF16
- Save Every: 500
- Max Step Saves: 20

TRAINING Section

- Batch Size: 1 (2 if VRAM > 16GB)
- Learning Rate: 0.0002
- Steps: 6000 *(mostly enough)*

ADVANCED Section

- Turn ON **Do Differential Guidance** *(doesn't affect training quality, just enables guidance)*

DATASET Section

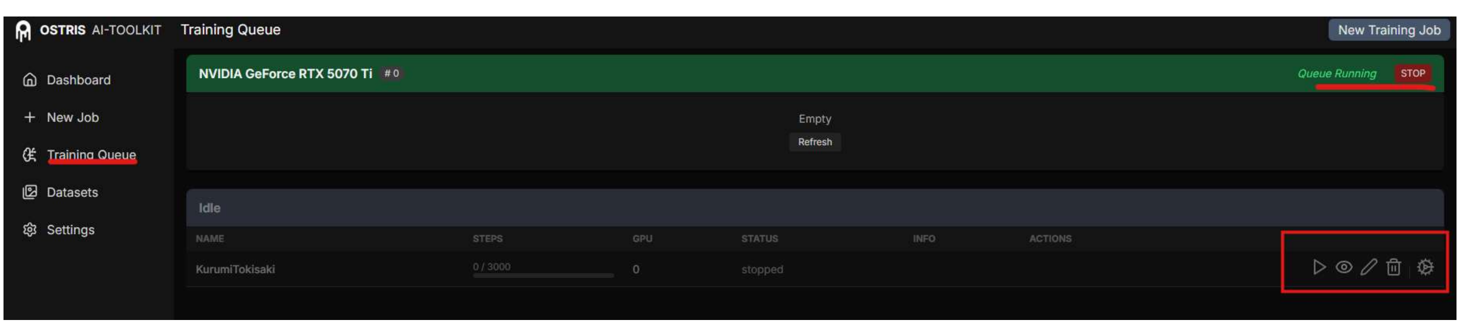
- Click **Add Dataset** → select your character's dataset
- Num Repeats: 1 (for 100+ images) / 2-3 (for ~50 images)

SAMPLE Section *(change)*

- Fill in your test prompt under **Sample Prompts** to monitor training progress visually

STEP 4: Start Training / Pause / Resume / Edit

Note: If you remove the character output folder from `\Ai-Toolkit\Output`, or delete important files, you will **not** be able to resume or edit training — it will simply restart from 0. You can, however, edit `.safetensors` files directly (e.g., `kurosaki-500.safetensors` or `kurosaki.safetensors`).



Training Locally and Enjoy Lora

-> Q/A

Q1: What tools can I use if I don't have enough VRAM?

A: If your GPU lacks sufficient VRAM, you have two excellent cloud-based alternatives. Use **RunPod** for LoRA and Checkpoint training, which provides powerful GPU instances on-demand. For running ComfyUI workflows in the cloud, use **GoogleColab** instead. You can get the Jupyter notebook for GoogleColab at my **Github**. Both options let you train models and generate images without investing in expensive hardware.

Note: Due to recent updates on CivitAI platforms, some file attachments may no longer be visible or accessible. For all resources related to this article — along with additional tools, files, and miscellaneous resources — please refer HERE